



Homework 9: This homework discusses the scattering amplitude $f(\theta)$ for a two-body system. Starting from the two-body Hamiltonian and separating it into center-of-mass and relative motion, we will see how this leads to outgoing spherical waves at large distances. The bonus questions explore a scattering problem with particles that are identical, and how the symmetrization or anti-symmetrization of the wavefunction enforces selection rules.

Take two particles of equal mass $m_1 = m_2 = m$ interacting through a central potential

$$V = V(|\mathbf{r}_1 - \mathbf{r}_2|), \quad (1)$$

The full Hamiltonian is

$$\hat{H} = \frac{\hat{\mathbf{p}}_1^2}{2m} + \frac{\hat{\mathbf{p}}_2^2}{2m} + V(|\mathbf{r}_1 - \mathbf{r}_2|). \quad (2)$$

We want to show that, in the center-of-mass frame, the wavefunction of relative motion takes the standard scattering form

$$\psi(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k} \cdot \mathbf{r}} + f(\theta) \frac{e^{ikr}}{r}. \quad (3)$$

a) Define center-of-mass and relative coordinates and their conjugate momenta:

$$\mathbf{R} \equiv \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2), \quad \mathbf{r} \equiv \mathbf{r}_1 - \mathbf{r}_2, \quad \mathbf{P} \equiv \mathbf{p}_1 + \mathbf{p}_2, \quad \mathbf{p} \equiv \frac{1}{2}(\mathbf{p}_1 - \mathbf{p}_2). \quad (4)$$

Verify that $[\mathbf{R}_i, \mathbf{P}_j] = i\delta_{ij}$ and $[\mathbf{r}_i, \mathbf{p}_j] = i\delta_{ij}$, while the mixed commutators $[\mathbf{R}_i, \mathbf{p}_j]$ and $[\mathbf{r}_i, \mathbf{P}_j]$ vanish. This shows that $(\mathbf{R}, \mathbf{P})$ and $(\mathbf{r}, \mathbf{p})$ are two independent canonical pairs.

b) Rewrite the kinetic energy of Eq. (2) in terms of $\mathbf{P}$ and $\mathbf{p}$, and show that the Hamiltonian looks like

$$\hat{H} = \frac{\hat{\mathbf{P}}^2}{2M} + \frac{\hat{\mathbf{p}}^2}{2\mu} + V(r), \quad \text{with} \quad M = 2m, \quad \mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m}{2}, \quad r = |\mathbf{r}|. \quad (5)$$

Here, the total mass is M and the reduced mass is μ . The central potential depends only on the relative coordinate by assumption.



This means that since $(\mathbf{R}, \mathbf{P})$ and $(\mathbf{r}, \mathbf{p})$ are canonical pairs, $\mathbf{P} = -i\nabla_{\mathbf{R}}$ and $\mathbf{p} = -i\nabla_{\mathbf{r}}$, so $\mathbf{P}^2 = -\nabla_{\mathbf{R}}^2$ and $\mathbf{p}^2 = -\nabla_{\mathbf{r}}^2$.

c) Because $\mathbf{P}$ commutes with $\hat{H}$, we can look for simultaneous eigenstates of $\hat{H}$ and $\mathbf{P}$. Substitute the wavefunction

$$\Psi(\mathbf{R}, \mathbf{r}) = e^{i\mathbf{K} \cdot \mathbf{R}} \psi(\mathbf{r}), \quad (6)$$

into Schrödinger's equation $\hat{H}|\Psi\rangle = E_{\text{tot}}|\Psi\rangle$ and show that $\psi(\mathbf{r})$ satisfies

$$\left[-\frac{\nabla_{\mathbf{r}}^2}{2\mu} + V(r) \right] \psi(\mathbf{r}) = E \psi(\mathbf{r}), \quad E \equiv E_{\text{tot}} - \frac{\mathbf{K}^2}{2M}. \quad (7)$$

Now we will be concerned with the center-of-mass frame, $\mathbf{K} = \mathbf{0}$, in which E is the total kinetic energy of relative motion and the lab and CM scattering angles coincide.

d) For a scattering state with energy $E = k^2/(2\mu)$, and for well outside the range of V , the function $\psi(\mathbf{r})$ satisfies the free Helmholtz equation

$$(\nabla^2 + k^2)\psi(\mathbf{r}) = 0. \quad (8)$$

Show that an incoming plane wave $e^{i\mathbf{k} \cdot \mathbf{r}}$ is a solution, and that the spherical wave e^{ikr}/r is also a solution. Explain in words why both pieces are needed in the asymptotic form Eq. (3). Why

does the scattering amplitude depends only on the zenith angle θ (the angle between $\hat{\mathbf{r}}$ and $\hat{\mathbf{k}}$) $f = f(\theta)$? Give at least one reason why $f(\theta)$ cannot be completely spherically symmetric, i.e., why f cannot be a constant independent of θ .

e) Compute the probability current

$$\mathbf{j} = \frac{1}{2\mu i} (\psi^* \nabla \psi - \psi \nabla \psi^*) \quad (9)$$

for the outgoing spherical wave $\psi_s(\mathbf{r}) = f(\theta) e^{ikr}/r$ alone, and show that the radial component at large r is

$$j_r^s = \frac{k}{\mu} \frac{|f(\theta)|^2}{r^2} + \mathcal{O}(r^{-3}). \quad (10)$$

Compare with the incident flux $j_i = k/\mu$ from the plane wave and give an expression for the differential cross section $d\sigma/d\Omega$ in terms of $f(\theta)$.



2) Bonus question on identical particles.

When the two scattering particles are identical, the detector at angle θ cannot tell which particle is which. Exchanging the two particles in the CM frame corresponds to $\mathbf{r} \rightarrow -\mathbf{r}$, which in spherical coordinates means $(\theta, \phi) \rightarrow (\pi - \theta, \phi + \pi)$. The total wavefunction must therefore be either symmetric (bosons) or antisymmetric (fermions) under this exchange.

a) Show that under $\mathbf{r} \rightarrow -\mathbf{r}$ the asymptotic form Eq. (3) becomes

$$\psi(-\mathbf{r}) \sim e^{-i\mathbf{k}\cdot\mathbf{r}} + f(\pi - \theta) \frac{e^{ikr}}{r}, \quad (11)$$

and construct the (anti)symmetrized spatial wavefunction

$$\psi_{\pm}(\mathbf{r}) = \frac{1}{\sqrt{2}} [\psi(\mathbf{r}) \pm \psi(-\mathbf{r})]. \quad (12)$$

Identify the symmetrized scattering amplitude $f_{\pm}(\theta) \equiv f(\theta) \pm f(\pi - \theta)$.

b) Because the two particles are indistinguishable, a detector at angle θ and a detector at angle $\pi - \theta$ count the same physical event. Taking proper account of this to avoid double-counting, the differential cross section is

$$\left. \frac{d\sigma}{d\Omega} \right|_{\pm} = |f(\theta) \pm f(\pi - \theta)|^2. \quad (13)$$

Expand this expression to show that it differs from the distinguishable-particle result by an *interference term*:

$$\left. \frac{d\sigma}{d\Omega} \right|_{\pm} = |f(\theta)|^2 + |f(\pi - \theta)|^2 \pm 2 \operatorname{Re}[f^*(\theta) f(\pi - \theta)]. \quad (14)$$

Show that at $\theta = \pi/2$ the symmetric (bosonic) cross section is twice the classical (distinguishable) result, while the antisymmetric (fermionic) cross section *vanishes*.

c) The scattering amplitude can be expanded in partial waves as

$$f(\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell + 1) e^{i\delta_{\ell}} \sin \delta_{\ell} P_{\ell}(\cos \theta), \quad (15)$$

where δ_{ℓ} are the phase shifts and P_{ℓ} are the Legendre polynomials. Using $P_{\ell}(\cos(\pi - \theta)) = P_{\ell}(-\cos \theta) = (-1)^{\ell} P_{\ell}(\cos \theta)$, show that

$$f(\theta) \pm f(\pi - \theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} [1 \pm (-1)^{\ell}] (2\ell + 1) e^{i\delta_{\ell}} \sin \delta_{\ell} P_{\ell}(\cos \theta). \quad (16)$$

What selection rule does this imply for the allowed partial waves ℓ in the scattering of identical bosons? How about identical fermions?